Measure Theoretic Probability
Lesson 21
Convergence in Probability

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Outline

1. Recap: Almost Sure Convergence
2. A Counterexample to the Converse of (3)
3. Convergence in Probability
4. An Example: $\xrightarrow{\mathbb{P}}$ but not $\xrightarrow{\text{a.s.}}$
5. $\xrightarrow{\text{a.s.}}$ Implies $\xrightarrow{\mathbb{P}}$
6. Summary and Exercises

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Where We Are

Setting

$X_1, X_2, \ldots$ and X are random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Random variables are not numbers, so to talk about $X_n \to X$ we must bring in real numbers through probabilities (or, later, expectations).

Four facts from the last two lessons

(1) **Definition.** $X_n \xrightarrow{\text{a.s.}} X$ iff $\mathbb{P}\big(\lim_n X_n = X\big) = \mathbb{P}\big(\{\omega : \lim_n X_n(\omega) = X(\omega)\}\big) = 1$.

(2) **Criterion.** $X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}\big(|X_n - X| > \varepsilon \text{ i.o.}\big) = 0$ for every $\varepsilon > 0$.

(3) **Sufficient.** $\sum_n \mathbb{P}(|X_n - X| > \varepsilon) < \infty$ for every $\varepsilon > 0 \;\Rightarrow\; X_n \xrightarrow{\text{a.s.}} X$.

(4) **Sufficient.** $\sum_n \mathbb{P}(|X_n - X| > \frac{1}{n}) < \infty \;\Rightarrow\; X_n \xrightarrow{\text{a.s.}} X$.

Question

(2) is an "if and only if"; (3) and (4) are one-directional. Whenever we see one-directional statements, we ask: **does the converse fail?**

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Goal and Construction

What we need

A sequence with $X_n \xrightarrow{\text{a.s.}} X$ for which $\sum_n \mathbb{P}(|X_n - X| > \varepsilon) = \infty$ for at least one $\varepsilon > 0$.

Example

$\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}([0,1])$, $\mathbb{P} =$ Lebesgue measure, and

$$X_n(\omega) = 2^n I_{[0,\,1/n]}(\omega), \qquad X(\omega) = 0.$$

A fixed ω: take $\omega = \frac{1}{3}$

$$X_1 = 2, \quad X_2 = 4, \quad X_3 = 8, \quad X_4 = 0, \quad X_5 = 0, \ \ldots$$

The interval $[0, 1/n]$ contains $\frac{1}{3}$ only for $n \leq 3$, after which the indicator is 0 forever.

Almost Sure Convergence Holds

Pointwise limit

Let $\omega \in (0, 1]$. Once $n > 1/\omega$, we have $\omega > 1/n$, so $X_n(\omega) = 0$ for all such n. Hence $X_n(\omega) \to 0 = X(\omega)$.

The exceptional point

At $\omega = 0$: $X_n(0) = 2^n \to \infty$, so there is no convergence there.

Conclusion

The convergence set is $A = (0, 1]$, with $\mathbb{P}(A) = 1$. So

$$X_n \xrightarrow{\text{a.s.}} 0.$$

Remark

$X \equiv 0$ is the "very boring" random variable that equals 0 with probability one.

But the Sum Diverges

Compute $\mathbb{P}(|X_n - X| > \varepsilon)$

Take $0 < \varepsilon < 2$. Since $X_n \geq 0$, $|X_n - X| = X_n$, and $2^n \geq 2 > \varepsilon$ for every $n \geq 1$. Therefore

$$\{X_n > \varepsilon\} = \{\omega : \ 2^n I_{[0,1/n]}(\omega) > \varepsilon\} = [0, 1/n], \qquad \mathbb{P}(X_n > \varepsilon) = \frac{1}{n}.$$

(For $n = 1$ the set is all of $[0, 1]$, with probability 1; for $n = 2$ it is $[0, \frac{1}{2}]$, and so on.)

The series

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \varepsilon) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Conclusion

$X_n \xrightarrow{\text{a.s.}} 0$, yet the sum in (3) is infinite. So the finiteness of the sum is **sufficient but not necessary** for almost sure convergence.

Remark

In fact the sum diverges for **every** $\varepsilon > 0$: once $2^n > \varepsilon$, the term is again $1/n$.

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Definition

Definition

X_n **converges in probability** to X if for every $\varepsilon > 0$

$$\lim_{n\to\infty} \mathbb{P}\big(|X_n - X| > \varepsilon\big) = 0.$$

We write $X_n \xrightarrow{\mathbb{P}} X$.

In set language

$\mathbb{P}\big(\{\omega : \ |X_n(\omega) - X(\omega)| > \varepsilon\}\big) \to 0$ as $n \to \infty$. This is a limit of **real numbers** in $[0, 1]$.

Compare with almost sure convergence

For $\xrightarrow{\text{a.s.}}$ the limit is **inside** the probability: $\mathbb{P}(\lim_n X_n = X) = 1$. For $\xrightarrow{\mathbb{P}}$ the limit is **outside**: $\lim_n \mathbb{P}(|X_n - X| > \varepsilon) = 0$. The second is much more concrete: compute probabilities, then take a limit of numbers.

Other name

In a general (non-probability) measure space, convergence in probability is called **convergence in measure**.

Equivalent Forms of the Definition

Theorem

For a sequence X_n and a random variable X, the following are equivalent:

(a) $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$;
(b) $\mathbb{P}(|X_n - X| \geq \varepsilon) \to 0$ for every $\varepsilon > 0$;
(c) $\mathbb{P}(|X_n - X| \leq \varepsilon) \to 1$ for every $\varepsilon > 0$;
(d) $\mathbb{P}(|X_n - X| < \varepsilon) \to 1$ for every $\varepsilon > 0$.

Remark

(a) $\iff$ (d) and (b) $\iff$ (c) are just complements, since $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$. The real content is (a) $\iff$ (b).

Proof: (b) $\Rightarrow$ (a)

Inclusion

If $|X_n(\omega) - X(\omega)| > \varepsilon$ then certainly $|X_n(\omega) - X(\omega)| \geq \varepsilon$:

$$\{|X_n - X| > \varepsilon\} \subseteq \{|X_n - X| \geq \varepsilon\}.$$

Monotonicity of $\mathbb{P}$

$$0 \leq \mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{P}(|X_n - X| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0.$$

By the squeeze principle, $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$. $\square$

Proof: (a) $\Rightarrow$ (b)

Inclusion with a smaller threshold

Fix $\varepsilon > 0$. If the distance is $\geq \varepsilon$, it is certainly $> \varepsilon/2$ (if a distance is at least 5, it is greater than 4):

$$\{|X_n - X| \geq \varepsilon\} \subseteq \{|X_n - X| > \tfrac{\varepsilon}{2}\}.$$

Monotonicity and limits

$$0 \leq \mathbb{P}(|X_n - X| \geq \varepsilon) \leq \mathbb{P}\big(|X_n - X| > \tfrac{\varepsilon}{2}\big) \xrightarrow[n\to\infty]{} 0,$$

because (a) holds for **every** positive number, in particular for $\varepsilon/2$. □

Remark

Any threshold in $(0, \varepsilon)$ would do; $\varepsilon/2$ is just a convenient choice that stays positive and strictly below ε.

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Construction: Moving Indicators

Setup

$\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}([0,1])$, $\mathbb{P} =$ Lebesgue. Let $X_n(\omega) = \omega + I_{[a_n,b_n]}(\omega)$, where the intervals $[a_n, b_n]$ are chosen in blocks:

n	Intervals $[a_n, b_n]$
1	$[0,1]$
$2,3$	$[0,\frac{1}{2}],\ [\frac{1}{2},1]$
$4,5,6$	$[0,\frac{1}{3}],\ [\frac{1}{3},\frac{2}{3}],\ [\frac{2}{3},1]$
$7,\ldots,10$	$[0,\frac{1}{4}],\ [\frac{1}{4},\frac{1}{2}],\ [\frac{1}{2},\frac{3}{4}],\ [\frac{3}{4},1]$
$\vdots$	$\vdots$

Block k consists of the k intervals of length $\frac{1}{k}$ that tile $[0,1]$. Overlapping endpoints do not matter.

Guess

The intervals shrink, so we expect $X_n \to X$ with $X(\omega) = \omega$ in some sense.

Not Almost Sure Convergence

A fixed point: $\omega = 0.28$

$$X_1 = 1.28, \quad X_2 = 1.28, \quad X_3 = 0.28, \quad X_4 = 1.28, \quad X_5 = 0.28, \ \ldots$$

The indicator **moves** instead of shrinking away from ω, so it keeps returning.

In general

In every block, ω lies in at least one interval, and (from block 2 on) outside at least one other. So $I_{[a_n, b_n]}(\omega) = 1$ infinitely often and $= 0$ infinitely often. Thus $X_n(\omega)$ does **not converge for any** $\omega \in [0, 1]$.

Conclusion

X_n does not converge almost surely to anything: the set where the sequence converges is empty.

But $X_n \xrightarrow{\mathbb{P}} X$ with $X(\omega) = \omega$

Compute

For $\varepsilon > 0$, $|X_n(\omega) - X(\omega)| = I_{[a_n, b_n]}(\omega)$, since the ω's cancel.

- $0 < \varepsilon < 1$: $\{I_{[a_n, b_n]} > \varepsilon\} = [a_n, b_n]$, so

$$\mathbb{P}(|X_n - X| > \varepsilon) = b_n - a_n \longrightarrow 0,$$

  because the interval lengths are $\frac{1}{k} \to 0$ as $n \to \infty$.
- $\varepsilon \geq 1$: the indicator is never greater than ε, so the set is $\varnothing$ and the probability is 0.

Conclusion

$\lim_n \mathbb{P}(|X_n - X| > \varepsilon) = 0$ for every $\varepsilon > 0$, so $X_n \xrightarrow{\mathbb{P}} X$, although X_n does **not** converge almost surely.

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Theorem and Proof Idea

Theorem

If $X_n \xrightarrow{\text{a.s.}} X$, then $X_n \xrightarrow{\mathbb{P}} X$. The converse is false (previous example).

Tools

- The criterion: $X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}(A_n^\varepsilon \text{ i.o.}) = 0$ for all $\varepsilon > 0$, where $A_n^\varepsilon = \{|X_n - X| > \varepsilon\}$ and $\{A_n^\varepsilon \text{ i.o.}\} = \limsup_n A_n^\varepsilon$.
- The inequality chain from the lessons on continuity of probabilities: for events A_n,

$$\mathbb{P}\big(\liminf_n A_n\big) \le \liminf_n \mathbb{P}(A_n) \le \limsup_n \mathbb{P}(A_n) \le \mathbb{P}\big(\limsup_n A_n\big).$$

Proof

Step 1

Fix $\varepsilon > 0$. By the criterion, $\mathbb{P}\big(\limsup_n A_n^\varepsilon\big) = 0$.

Step 2: squeeze

Apply the chain to $A_n = A_n^\varepsilon$:

$$0 \le \liminf_n \mathbb{P}(A_n^\varepsilon) \le \limsup_n \mathbb{P}(A_n^\varepsilon) \le \mathbb{P}\big(\limsup_n A_n^\varepsilon\big) = 0.$$

(Probabilities are nonnegative, so the left side is ≥ 0.) Hence $\liminf_n \mathbb{P}(A_n^\varepsilon) = \limsup_n \mathbb{P}(A_n^\varepsilon) = 0$.

Step 3

When the lim inf and lim sup of a real sequence agree, the limit exists and equals the common value. So

$$\lim_{n\to\infty} \mathbb{P}\big(|X_n - X| > \varepsilon\big) = 0 \quad \text{for every } \varepsilon > 0,$$

that is, $X_n \xrightarrow{\mathbb{P}} X$. □

Strong and weak

Almost sure convergence is the **stronger** mode, and convergence in probability is the **weaker** mode.

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Summary

Key results

Concept / Result	Content
Counterexample to converse of (3)	$X_n = 2^n I_{[0,1/n]} \xrightarrow{\text{a.s.}} 0$ but $\sum \mathbb{P}(X_n > \varepsilon) = \sum 1/n = \infty$
Convergence in probability	$\lim_n \mathbb{P}(\lvert X_n - X\rvert > \varepsilon) = 0$ for all $\varepsilon > 0$ (limit outside $\mathbb{P}$)
Equivalent forms	$> \varepsilon$, $\geq \varepsilon$, and the complements $< \varepsilon$, $\leq \varepsilon$ with limit 1
Moving indicators	$X_n = \omega + I_{[a_n,b_n]}$: $\xrightarrow{\mathbb{P}}$ ω but converges at **no** ω
$\xrightarrow{\text{a.s.}} \Rightarrow \xrightarrow{\mathbb{P}}$	via $\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n) \leq \limsup \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n)$
Converse	$\xrightarrow{\mathbb{P}} \not\Rightarrow \xrightarrow{\text{a.s.}}$

Exercises

1. For $X_n = 2^n I_{[0,1/n]}$, verify that $\{X_n > \varepsilon\} = [0, 1/n]$ for $0 < \varepsilon < 2$. What is $\{X_n > \varepsilon\}$ for $\varepsilon \geq 2$, and does $\sum_n \mathbb{P}(X_n > \varepsilon)$ converge there?
2. Show directly that $X_n = 2^n I_{[0,1/n]} \xrightarrow{\mathbb{P}} 0$.
3. Prove that (c) and (d) in the equivalence theorem are equivalent to (a) by taking complements.
4. In the moving-indicator example, find explicit formulas for a_n, b_n in terms of the block index k and the position j within the block.
5. Show that in the moving-indicator example $\sum_n \mathbb{P}(|X_n - X| > \varepsilon) = \infty$ for $0 < \varepsilon < 1$. Is this consistent with fact (3)?
6. Prove the chain $\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n) \leq \limsup \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n)$.
7. Show that if $X_n \xrightarrow{\mathbb{P}} X$ and $X_n \xrightarrow{\mathbb{P}} Y$, then $\mathbb{P}(X \neq Y) = 0$. (Hint: $\{|X - Y| > 2\varepsilon\} \subseteq \{|X_n - X| > \varepsilon\} \cup \{|X_n - Y| > \varepsilon\}$.)

Outlook

Next lesson

Subsequences. If a real sequence $a_n \to a$, every subsequence converges to a. The same holds for $\xrightarrow{\text{a.s.}}$ and $\xrightarrow{\mathbb{P}}$. Conversely, a real sequence need not converge but may have a convergent subsequence (Bolzano–Weierstrass: bounded sequences have convergent subsequences). We study the analogous statements for $\xrightarrow{\text{a.s.}}$ and $\xrightarrow{\mathbb{P}}$.

Looking ahead

After that we define **expectation**, giving a third way to bring real numbers into the study of convergence.